Linux Fundamentals

1. pwd: This command is used to print the current working directory path.
2. ls: It lists all the files inside the directory.
3. ls -la: It prints the all the files inside the current directory. ‘l’ flag provides the detailed information view of the each files and ‘a’ flag is used for the hidden files in the directory.
4. cd ~: cd is used for the changing the directory. ~ is used to change the current directory to the home directory.
5. mkdir: Used for the creating a new directory . ‘p’ flag : If the directory already exists, if does not throw an error.
   For example: mkdir -p directory_name
6. cd dir_name: Used to navigate inside the particular directory.
7. touch file1.txt: touch command is used to create a empty file in current directory, if it does not exist.
8. echo "Hello Linux" > file1.txt: It writes the “Hello Linux” text to the file1.txt. If file does not exist, it is created.
9. cat file1.txt: cat command is used for printing the contents of the file to the terminal.
10. clear: clear is used to clear the entire terminal.
11. cp file1.txt file2.txt: cp command is used to copy the contents of the first file to the second. Here the contents of the file1.txt will be copied in the file2.txt.
12. mv file1.txt new.txt: Basically, it has 2 purposes. First, it renames the file1.txt to new.txt(Same folder, different name). Second, it moves the current file to the different folder.
13. rm new.txt: It removes the new.txt from the directory.
14. whoami: It prints the current user’s username to the terminal.
15. uptime: It prints the current system time, how long the system has been running, number of users logged in, the load average.
16. ps -A: It lists all the active processes on the system.
17. df -h: This command is used to display the disk usage of the filesystem. ‘-h’ flag is used for human-readable format.
18. free -h: It used to display the memory usage. It displays the total, used and free.

19. history: It displays the number of commands executed in the session.
20. sudo apt update: This is a Linux package management command. It updates the local package index from configured repositories.
21. sudo apt upgrade -y: It helps to upgrades installed packages to their latest available versions.
22. which: It shows the location of the executables. For example: which python3.

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purvaa@Purva: ~/linux_funda
purvaa@Purva:~$ pwd
/home/purvaa
purvaa@Purva:~$ ls
Deploysafe-gitops  minikube-linux-amd64  scripts  sync-jenkins-kubeconfig.sh  wget-log
purvaa@Purva:~$ ls -la
total 131876
drwxr-x--- 15 purvaa purvaa      4096 Jul 12 13:03 .
drwxr-xr-x  4 root   root        4096 Jun 24 17:13 ..
drwxr-xr-x  3 purvaa purvaa      4096 Jun 20 15:56 .ansible
lrwxrwxrwx  1 purvaa purvaa        24 Jun 14 22:19 .aws -> /mnt/c/Users/LENOVO/.aws
lrwxrwxrwx  1 purvaa purvaa        26 Jun 14 22:19 .azure -> /mnt/c/Users/LENOVO/.azure
-rw-------  1 purvaa purvaa     41886 Jul 13 11:10 .bash_history
-rw-r--r--  1 purvaa purvaa       220 Jun 14 08:32 .bash_logout
-rw-r--r--  1 purvaa purvaa      3771 Jun 14 08:32 .bashrc
drwxr-x---  7 purvaa purvaa      4096 Jul  2 17:32 .cache
drwxr-x---  4 purvaa purvaa      4096 Jun 17 12:38 .config
drwxr-xr-x  6 purvaa purvaa      4096 Jun 27 16:40 .docker
-rw-r--r--  1 purvaa purvaa        53 Jun 14 10:47 .gitconfig
drwx------  2 purvaa purvaa      4096 Jun 14 13:17 .gnupg
drwxr-xr-x  3 purvaa purvaa      4096 Jun 26 22:24 .kube
drwxr-xr-x  2 purvaa purvaa      4096 Jun 15 14:00 .landscape
-rw-------  1 purvaa purvaa        20 Jun 14 10:54 .lesshst
drwxr-xr-x 10 purvaa purvaa      4096 Jul  5 10:48 .minikube
-rw-rw-r--  1 purvaa purvaa         0 Jul 13 10:30 .motd_shown
drwxr-xr-x  4 purvaa purvaa      4096 Jun 14 13:11 .npm
-rw-r--r--  1 purvaa purvaa       807 Jun 14 08:32 .profile
drwx------  3 purvaa purvaa      4096 Jul  2 18:29 .ssh
drwxr-xr-x  2 purvaa purvaa      4096 Jul 10 17:09 .terraform.d
-rw-------  1 purvaa purvaa       923 Jul  1 12:06 .viminfo
-rw-r--r--  1 purvaa purvaa       164 Jun 14 23:24 .wget-hsts
drwxr-xr-x  5 purvaa purvaa      4096 Jul 12 13:03 Deploysafe-gitops
-rw-r--r--  1 purvaa purvaa 134891912 Jun 14 22:51 minikube-linux-amd64
drwxr-xr-x  2 purvaa purvaa      4096 Jul  1 12:05 scripts
-rwxr-xr-x  1 purvaa purvaa       391 Jul  1 12:06 sync-jenkins-kubeconfig.sh
-rw-r--r--  1 root   root         581 Jun 14 23:22 wget-log
purvaa@Purva:~$ mkdir linux_fundamentals
purvaa@Purva:~$ cd linux_fundamentals
purvaa@Purva:~/linux_fundamentals$ cd ~
purvaa@Purva:~$ cd linux_fundamentals
purvaa@Purva:~/linux_fundamentals$ touch file1.txt
purvaa@Purva:~/linux_fundamentals$ echo "Hello Linux" > file1.txt
purvaa@Purva:~/linux_fundamentals$ cat file1.txt
Hello Linux
purvaa@Purva:~/linux_fundamentals$
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purvaa@Purva: ~/linux_funda
purvaa@Purva:~/linux_fundamentals$ cp file1.txt file2.txt
purvaa@Purva:~/linux_fundamentals$ cat file2.txt
Hello Linux
purvaa@Purva:~/linux_fundamentals$ mv file1.txt new.txt
purvaa@Purva:~/linux_fundamentals$ ls
file2.txt  new.txt
purvaa@Purva:~/linux_fundamentals$ rm new.txt
purvaa@Purva:~/linux_fundamentals$ ls
file2.txt
purvaa@Purva:~/linux_fundamentals$ whoami
purvaa
purvaa@Purva:~/linux_fundamentals$ uptime
 11:54:20 up 4 min,  1 user,  load average: 1.74, 2.06, 1.05
purvaa@Purva:~/linux_fundamentals$ ps -A
    PID TTY          TIME CMD
      1 ?        00:00:02 systemd
      2 hvc0     00:00:00 init-systemd(Ub
      7 hvc0     00:00:00 init
     47 ?        00:00:00 systemd-journal
     79 ?        00:00:00 systemd-resolve
     90 ?        00:00:00 systemd-udevd
    150 ?        00:00:00 chronyd-starter
    151 ?        00:00:00 cron
    152 ?        00:00:00 dbus-daemon
    155 ?        00:00:46 java
    167 ?        00:00:00 networkd-dispat
    171 ?        00:00:00 systemd-logind
    174 ?        00:00:00 wsl-pro-service
    184 ?        00:01:39 k3s-server
    224 tty1     00:00:00 agetty
    230 ?        00:00:00 rsyslogd
    250 ?        00:00:00 unattended-upgr
    251 ?        00:00:00 chronyd
    273 ?        00:00:00 chronyd
    278 ?        00:00:01 containerd
    372 ?        00:00:00 dockerd
    656 ?        00:00:16 containerd
    829 ?        00:00:00 SessionLeader
    830 ?        00:00:00 Relay(833)
    833 pts/0    00:00:00 bash
    834 ?        00:00:00 login
    880 ?        00:00:00 systemd
    882 ?        00:00:00 (sd-pam)
    910 pts/1    00:00:00 bash
```

purvaa@Purva: ~/linux_funda

```
   6414 ?        00:00:11 python
   6464 ?        00:00:02 uvicorn
   6469 ?        00:00:02 uvicorn
   6478 ?        00:00:02 uvicorn
   6512 ?        00:00:00 tini
   6513 ?        00:00:08 argocd-applicat
   6526 ?        00:00:22 prometheus
   6529 ?        00:00:02 uvicorn
   6617 ?        00:00:03 argocd-repo-ser
   6728 ?        00:00:10 python
   6737 ?        00:00:01 alertmanager
   6818 ?        00:00:00 gpg-agent
   6865 ?        00:00:00 prometheus-conf
   6874 ?        00:00:15 grafana
   6889 ?        00:00:00 prometheus-conf
   7491 pts/0    00:00:00 ps
purvaa@Purva:~/linux_fundamentals$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G     0  1.9G   0% /usr/lib/modules/6.18.33.1-microsoft-standard-WSL2
none            1.9G  4.0K  1.9G   1% /mnt/wsl
drivers         231G  118G  114G  51% /usr/lib/wsl/drivers
/dev/sdd       1007G   16G  940G   2% /
none            1.9G   36K  1.9G   1% /mnt/wslg
none            1.9G     0  1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G  2.8M  1.9G   1% /init
none            1.9G  4.9M  1.9G   1% /run
none            1.9G     0  1.9G   0% /run/lock
none            1.9G     0  1.9G   0% /run/shm
none            1.9G   80K  1.9G   1% /mnt/wslg/versions.txt
none            1.9G   80K  1.9G   1% /mnt/wslg/doc
C:\             231G  118G  114G  51% /mnt/c
D:\             246G   13G  233G   6% /mnt/d
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.9G  4.8M  1.9G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           381M   12K  381M   1% /run/user/1000
purvaa@Purva:~/linux_fundamentals$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       2.7Gi        66Mi       1.1Mi       1.0Gi       992Mi
Swap:          1.0Gi       898Mi       125Mi
purvaa@Purva:~/linux_fundamentals$ which python3
/usr/bin/python3
purvaa@Purva:~/linux_fundamentals$ |
```